The Random Universe: a meeting to celebrate Andrew Jaffe at 60.

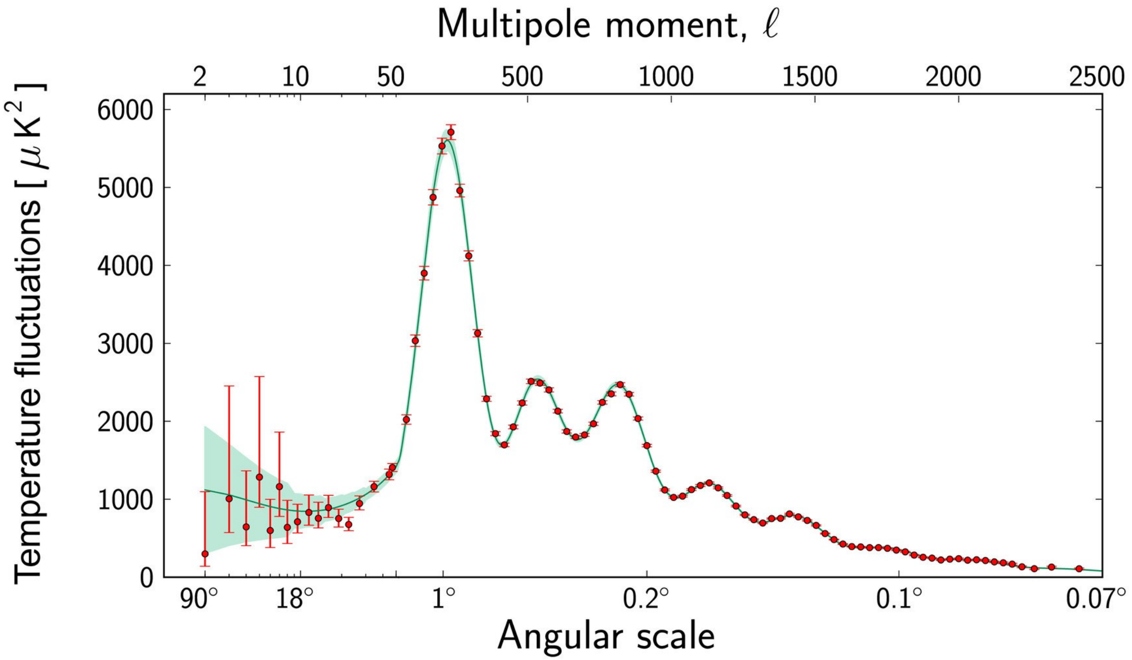
Strong non-Gaussianity of Planck polarization data and an alternative way to measure the CMB anisotropy and polarization at low multipoles

Dmitry Novikov, Imperial College, 29.06.2026.

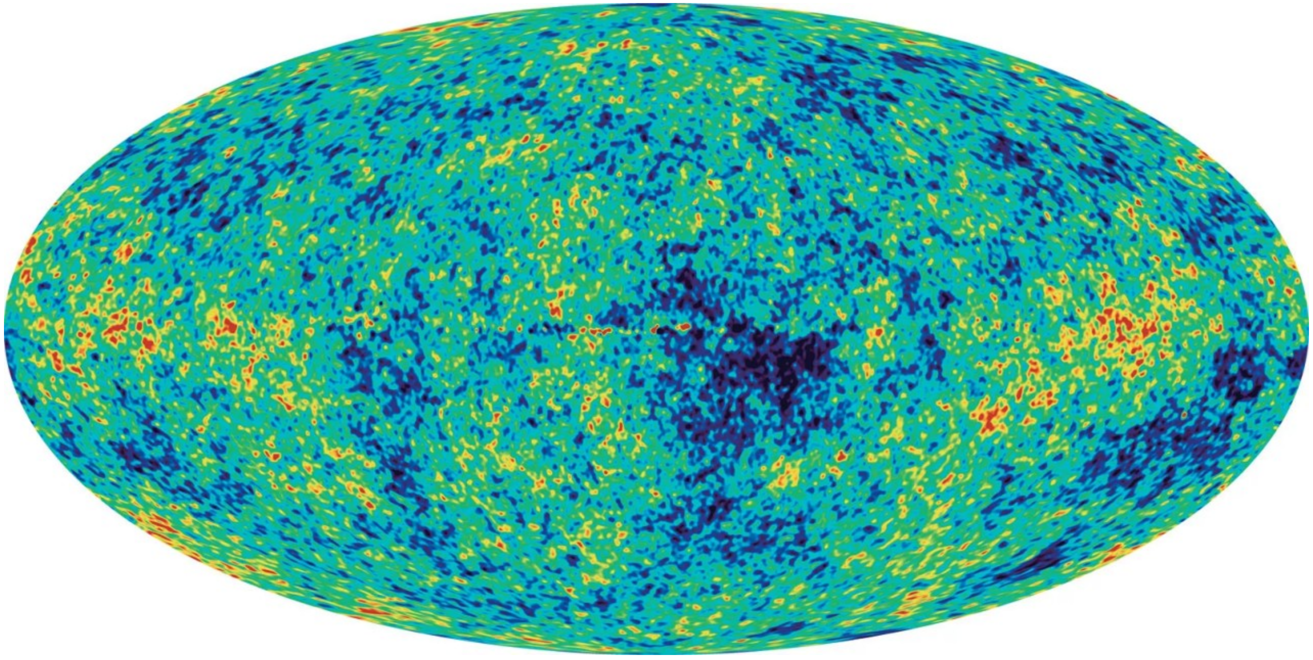
The Random Universe: a meeting to celebrate Andrew Jaffe at 60.

Why do we need a Gaussianity test?

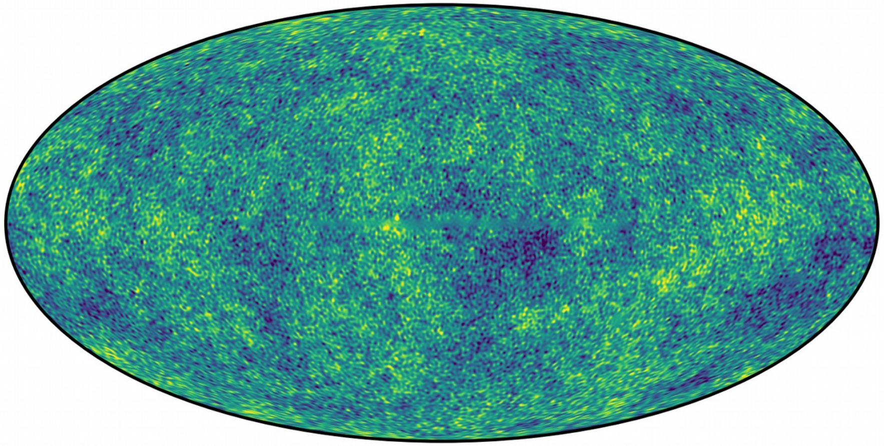
- Inflation predicts Gaussianity of initial perturbations
- General Gaussianity of cosmological E and B modes (in the first approximation)
- Initial non-Gaussianity (refinement of the inflation model)
- Presence of unremoved non-Gaussian foregrounds and distortions due to gravitational lensing (data cleaning)



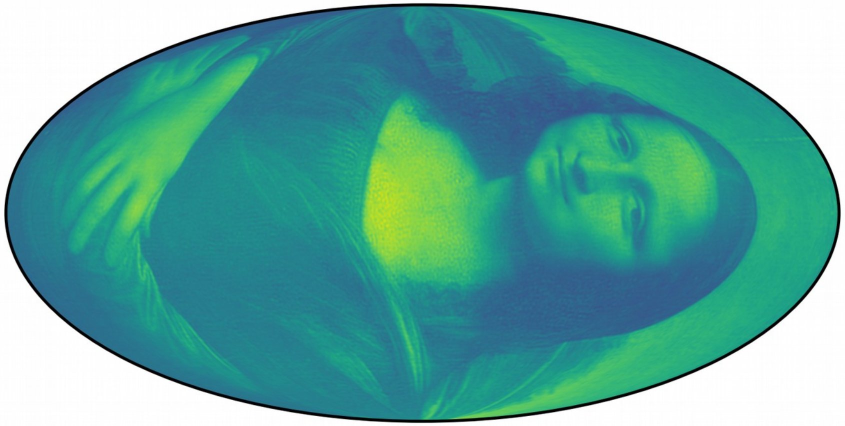
Bennett et al 2013 WMAP



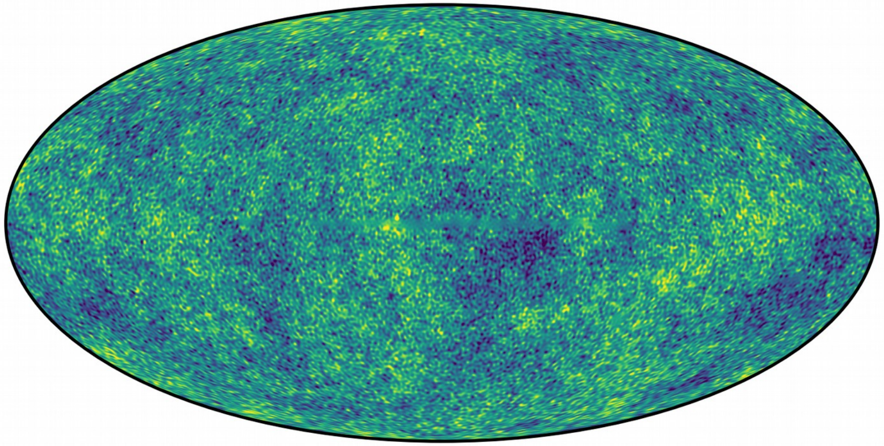
Planck 2018 results



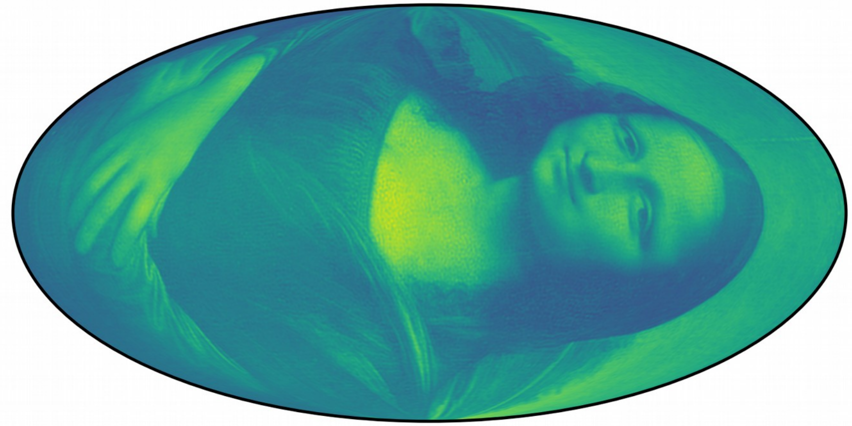
CMB spectrum, CMB phases



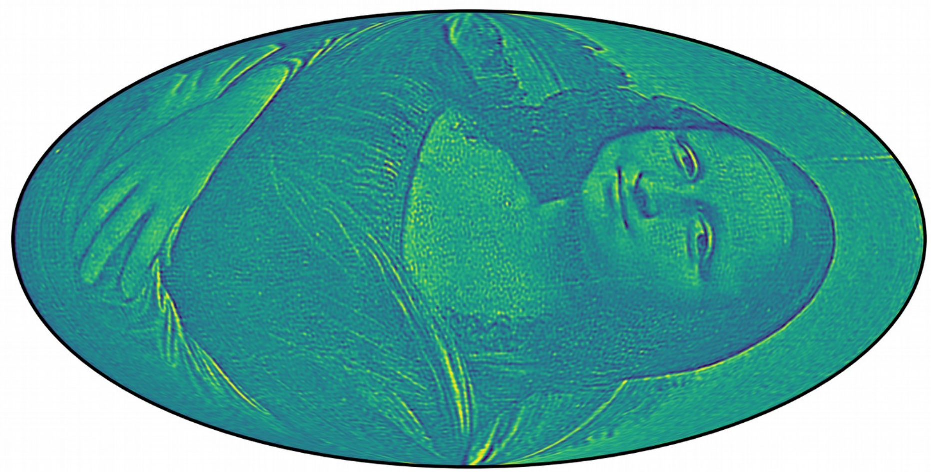
Mona Lisa spectrum, Mona Lisa phases



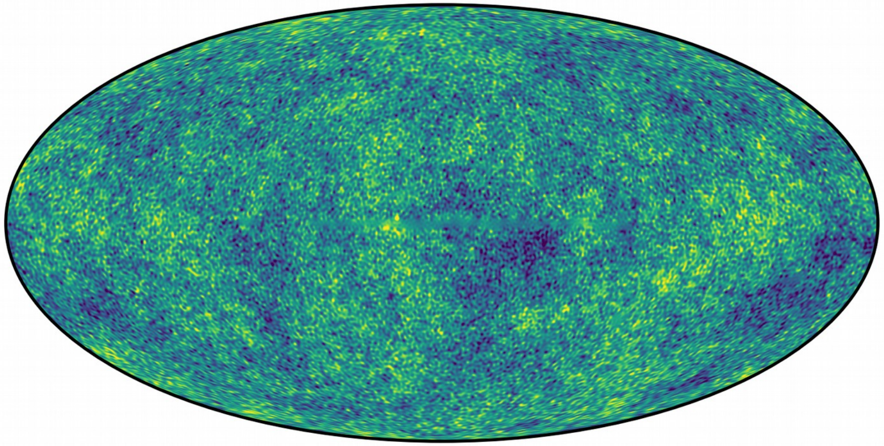
CMB spectrum, CMB phases



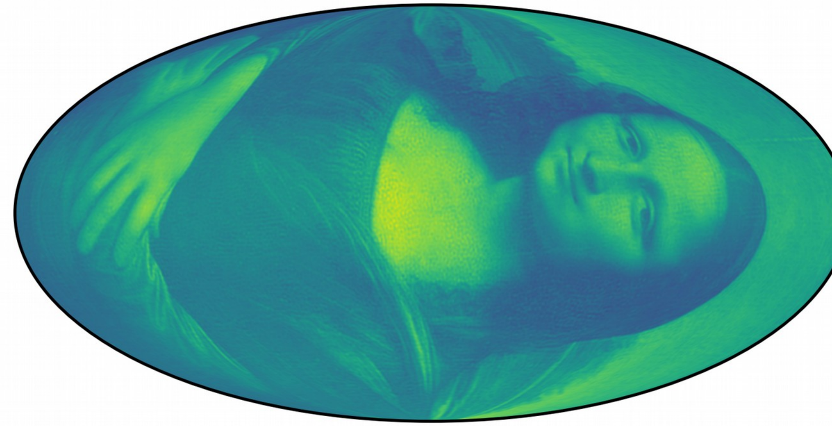
Mona Lisa spectrum, Mona Lisa phases



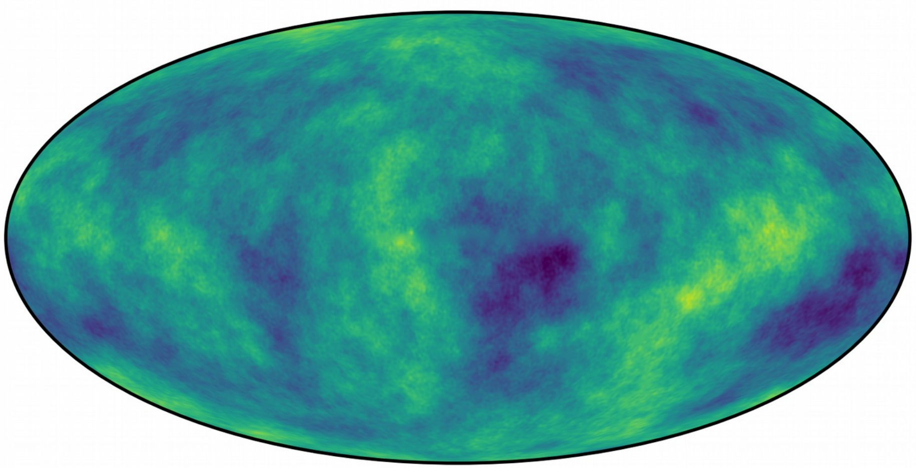
CMB spectrum, Mona Lisa phases



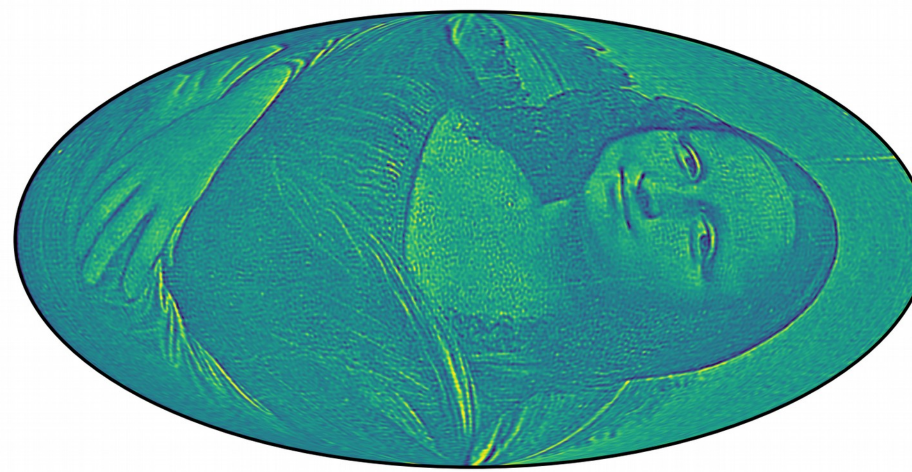
CMB spectrum, CMB phases



Mona Lisa spectrum, Mona Lisa phases



Mona Lisa spectrum, CMB phases

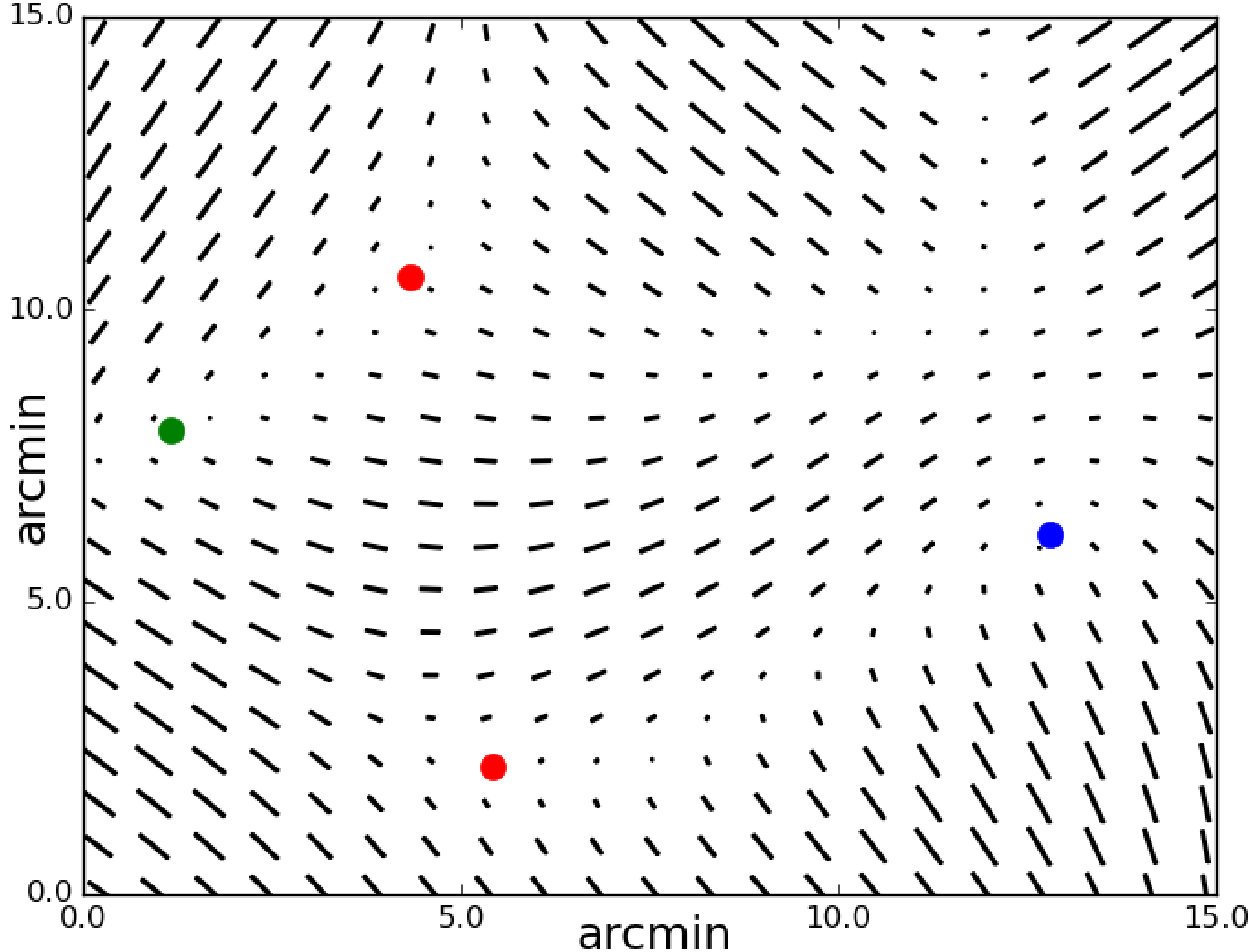


CMB spectrum, Mona Lisa phases

How to provide Gaussianity test for linear polarization?

Unpolarized points:

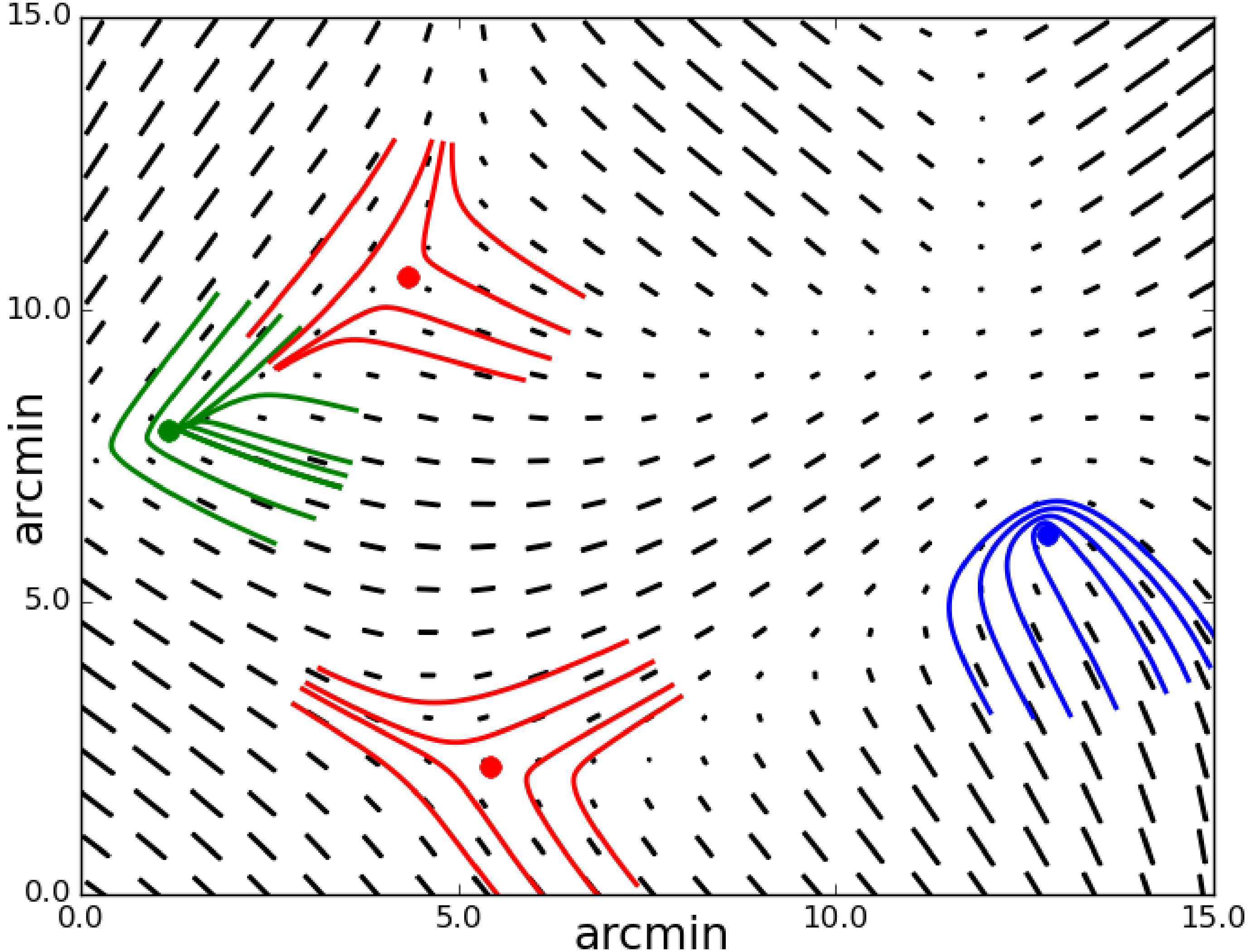
$$p = \sqrt{q^2 + u^2}, \quad \tan 2\varphi = \frac{u}{q}$$



$$q(x_0, y_0) = u(x_0, y_0) = 0 \quad => \quad p(x_0, y_0) = 0$$

Unpolarized points:

$$p = \sqrt{q^2 + u^2}, \quad \tan 2\varphi = \frac{u}{q}$$



$$q(x_0, y_0) = u(x_0, y_0) = 0 \quad => \quad p(x_0, y_0) = 0$$

3 types of unpolarized points:

$$p(x_0, y_0) = 0, \;\; \Delta x = x - x_0, \;\; \Delta y = y - y_0$$

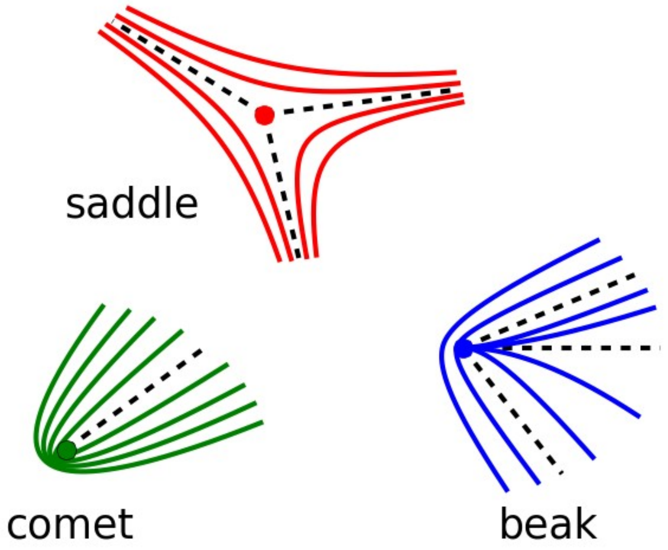
$$\begin{pmatrix} q \\ u \end{pmatrix} = \frac{\sigma_1}{\sigma_0} \cdot \begin{pmatrix} q_x & q_y \\ u_x & u_y \end{pmatrix} \cdot \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$$

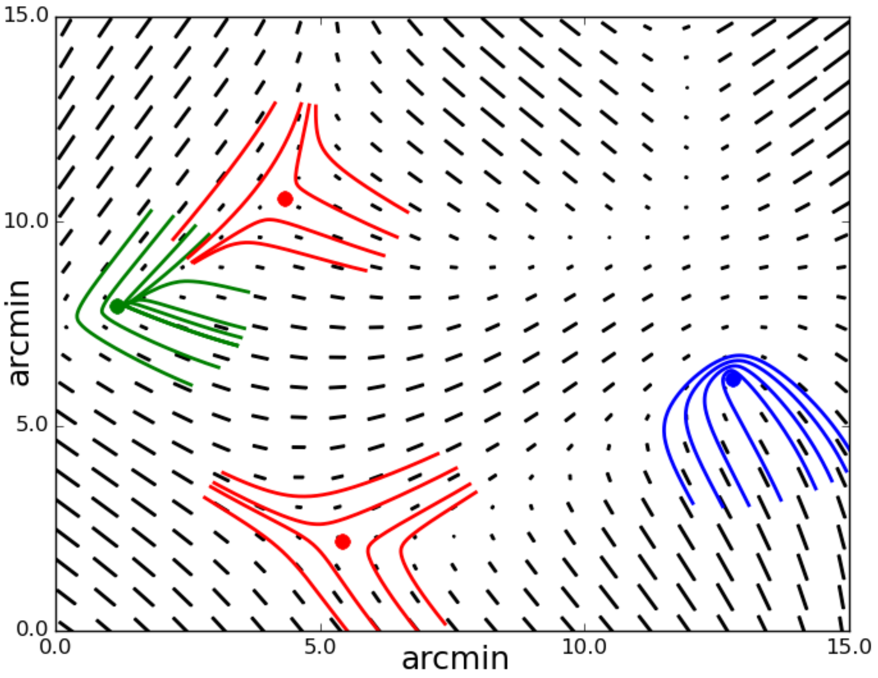
$$\begin{aligned} & d = q_x u_y - u_x q_y, \\ & D = 18abce - 4b^3 e + b^2 c^2 - 4ac^3 - 27a^2 e^2, \\ & a = u_y, \quad b = (u_x + 2q_y), \\ & c = (2q_x - u_y), \quad e = -u_x \end{aligned}$$

$$\begin{aligned} & d < 0, && \rightarrow \;\; saddle, \\ & d > 0, D > 0 && \rightarrow \;\; beak, \\ & d > 0, D < 0 && \rightarrow \;\; comet \end{aligned}$$

A. Dolgov, A. Doroshkevich, D. Novikov, and I.Novikov, IJMP D 8, 189 (1999), astro-ph/9901399

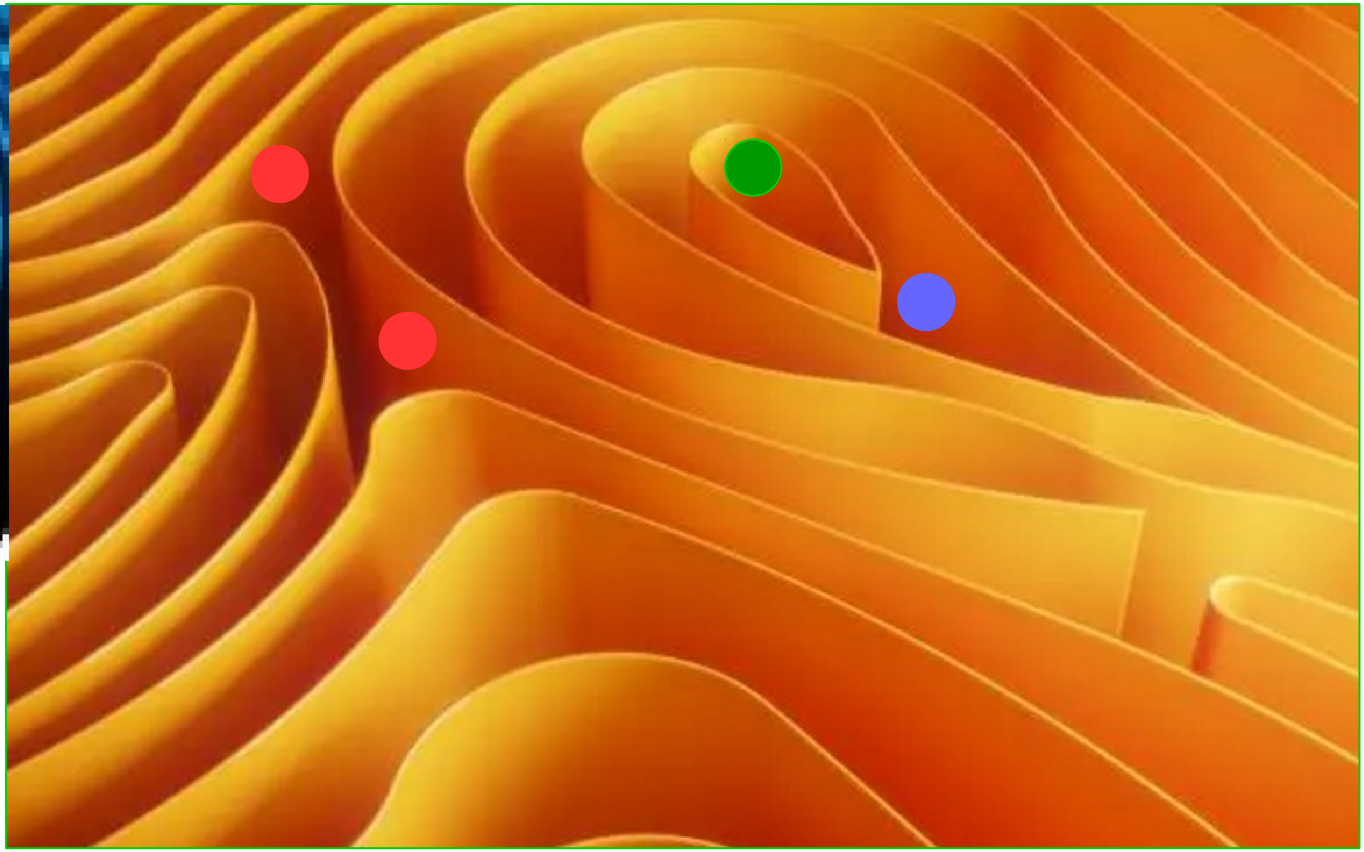
Unpolarized points





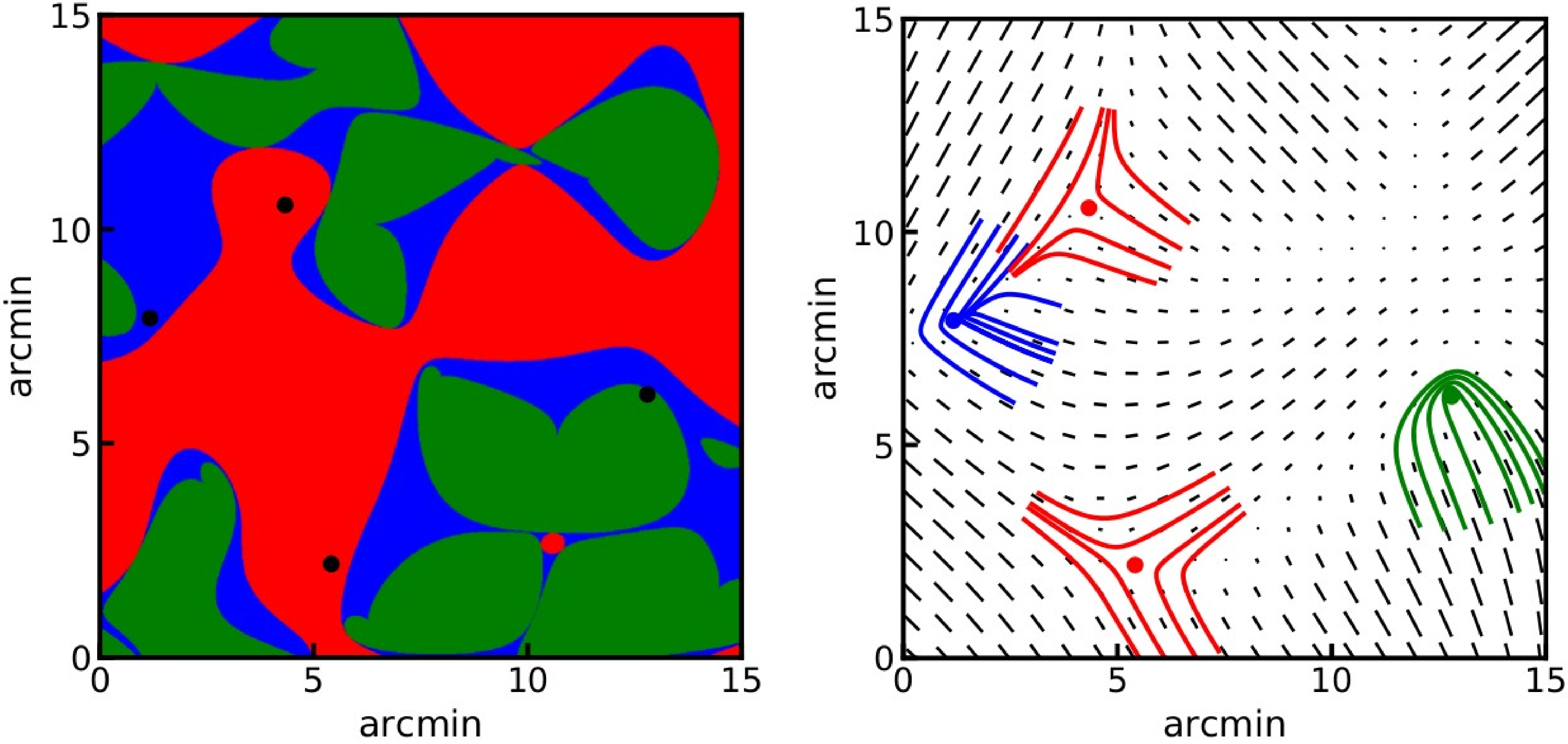
Fingerprints:

Fingerprints:



●	saddle
●	comet
●	beak

Area fractions and unpolarized points:



Distribution of unpolarized points (Gaussian field):

$Number\ of\ points:$

$\langle N_{tot} \rangle = \frac{A}{2\pi^2 r_c^2}$

$\langle N_s \rangle = 0.5 \cdot \langle N_{tot} \rangle$

$\langle N_c \rangle = 0.448 \cdot \langle N_{tot} \rangle$

$\langle N_b \rangle = 0.052 \cdot \langle N_{tot} \rangle$

$Area\ fraction:$

$A_{tot} = A,$

$\langle A_s \rangle = 0.5 \cdot A_{tot}$

$\langle A_c \rangle = 0.384 \cdot A_{tot}$

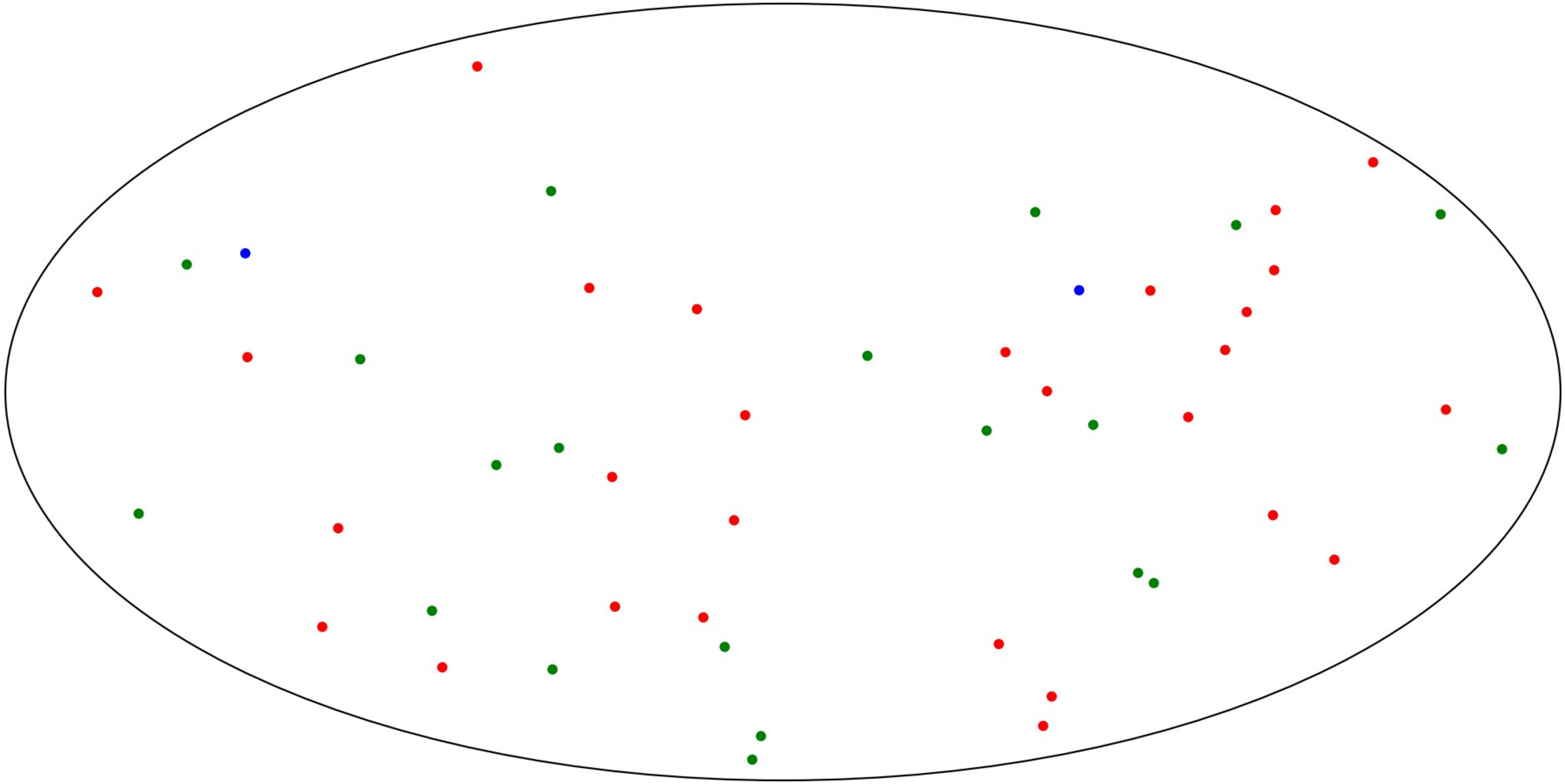
$\langle A_b \rangle = 0.116 \cdot A_{tot}$

Testing different angular scales for Gaussianity

$$E, B = \sum_{\ell=2}^{\ell_{max}} \sum_{m=-\ell}^{\ell} a_{\ell m}^{E,B} \cdot Y_{\ell m}(\theta, \varphi)$$

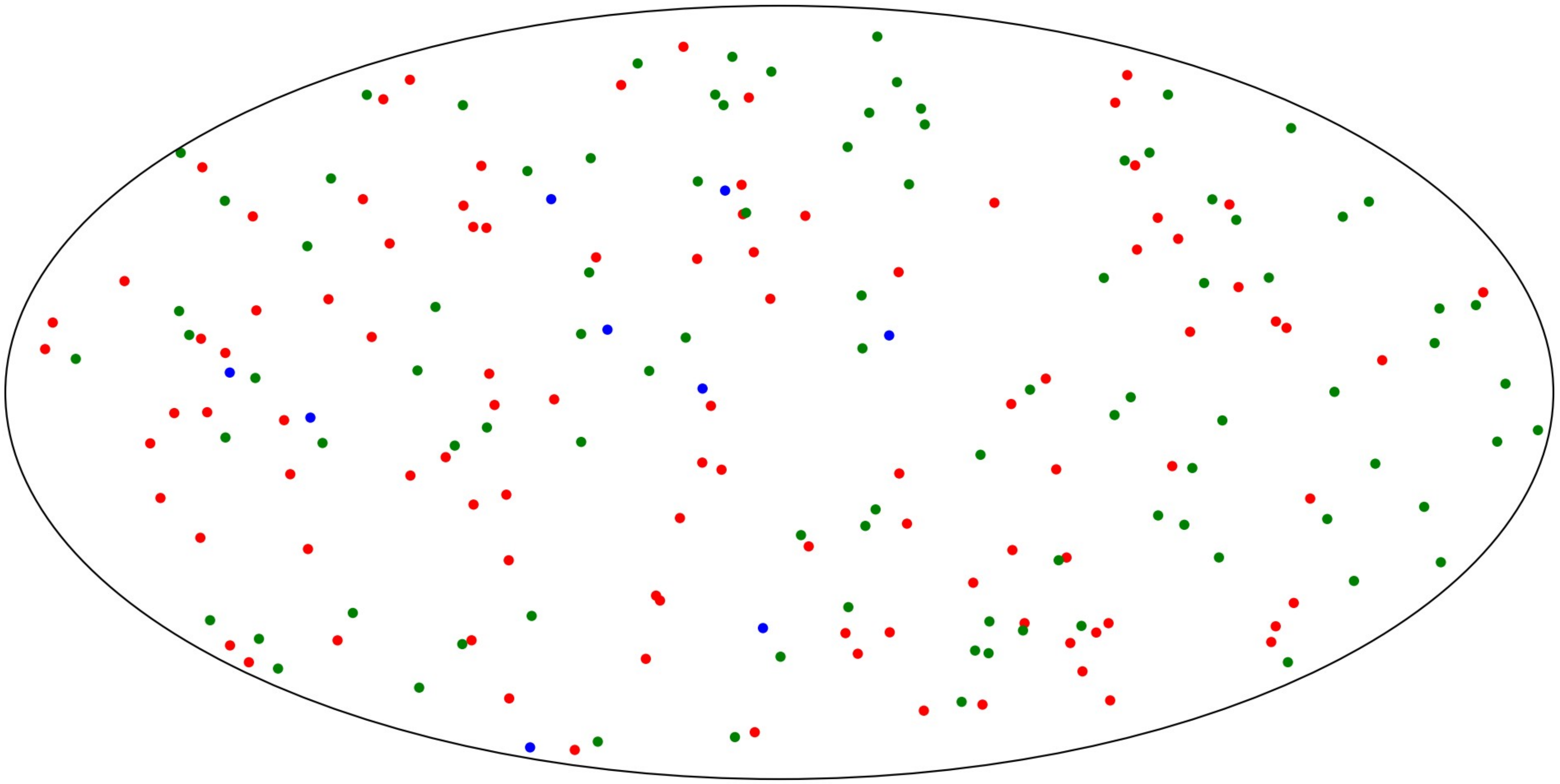
By changing the number of harmonics ℓ_{max} we test different angular scales

Distribution of unpolarized points (Gaussian field):



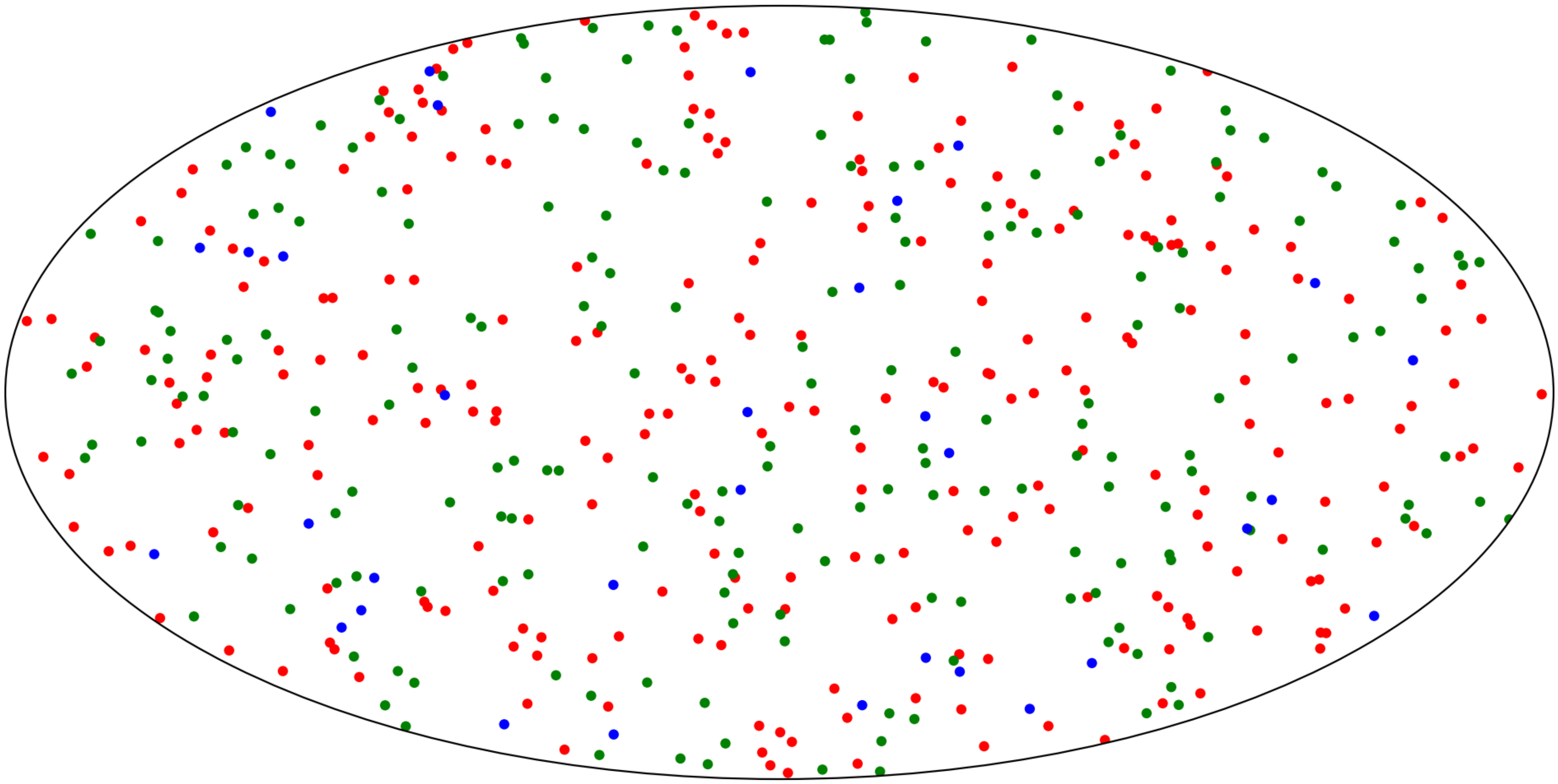
$$\ell_{max} = 7$$

Distribution of unpolarized points (Gaussian field):



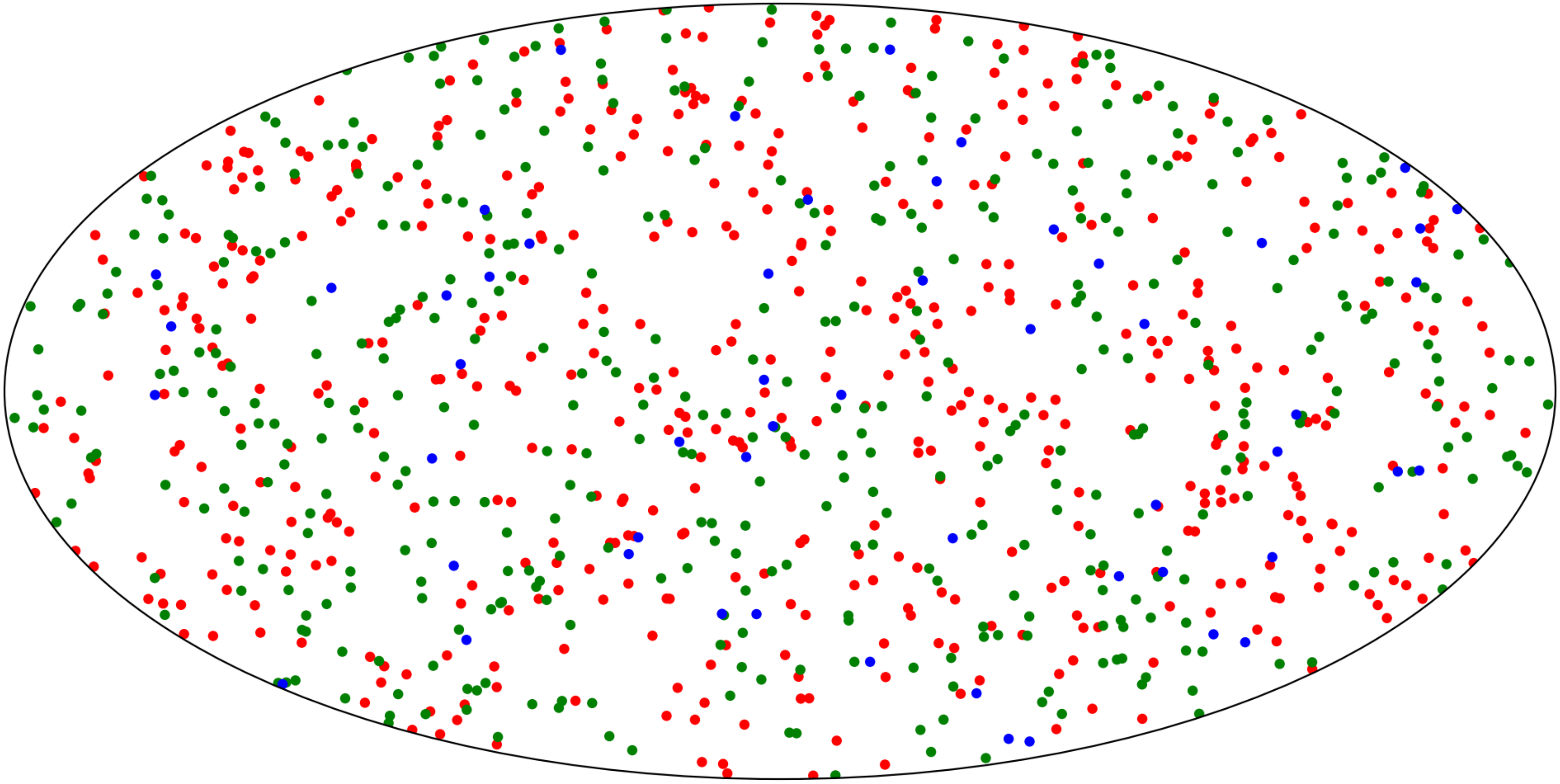
$$\ell_{max} = 11$$

Distribution of unpolarized points (Gaussian field):



$$\ell_{max} = 15$$

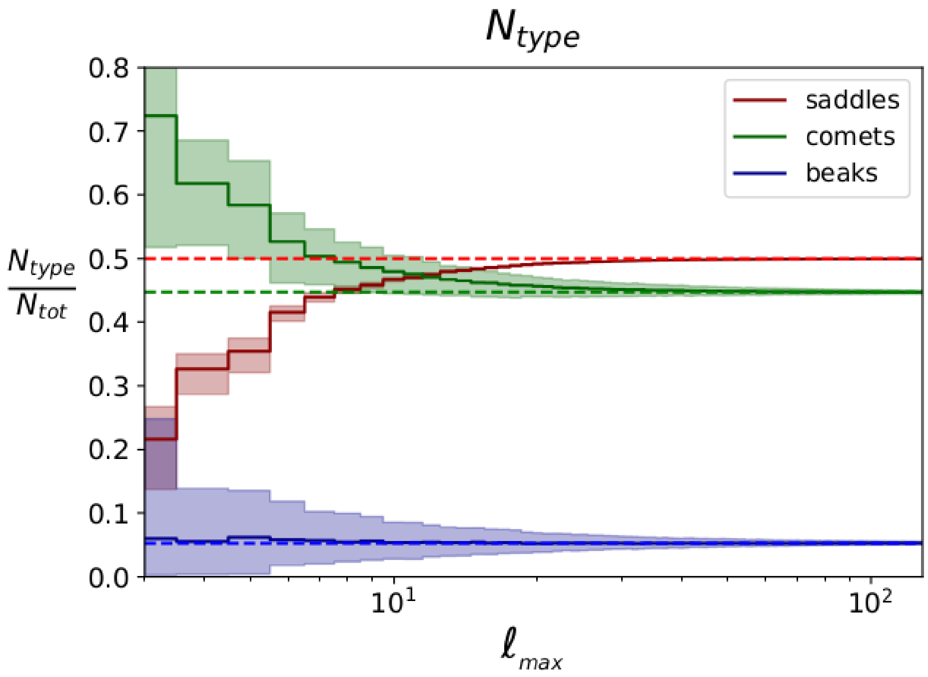
Distribution of unpolarized points (Gaussian field):



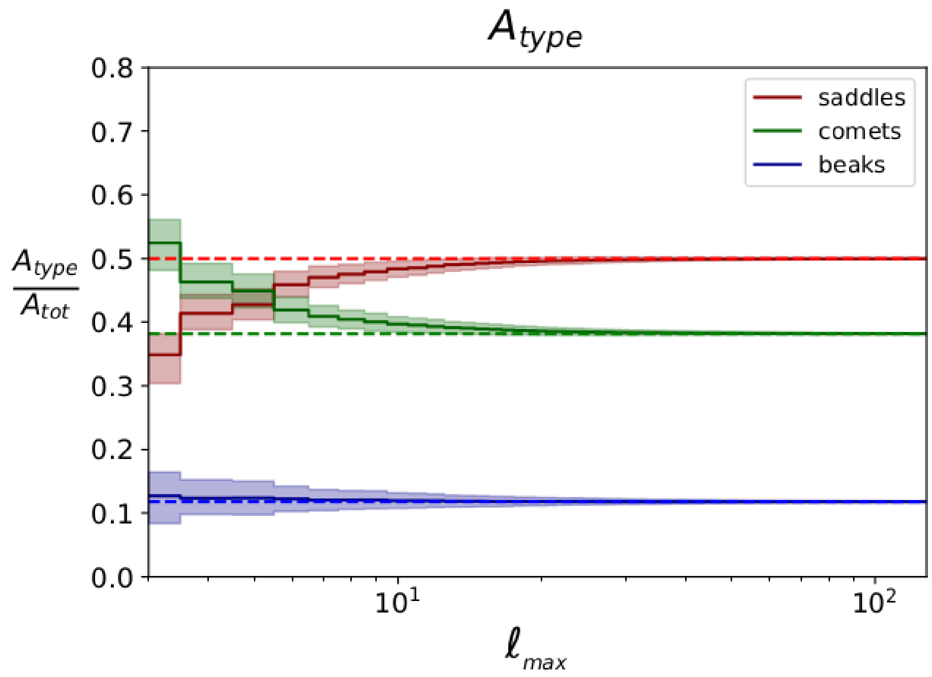
$$\ell_{max} = 23$$

Distribution of unpolarized points (Gaussian field):

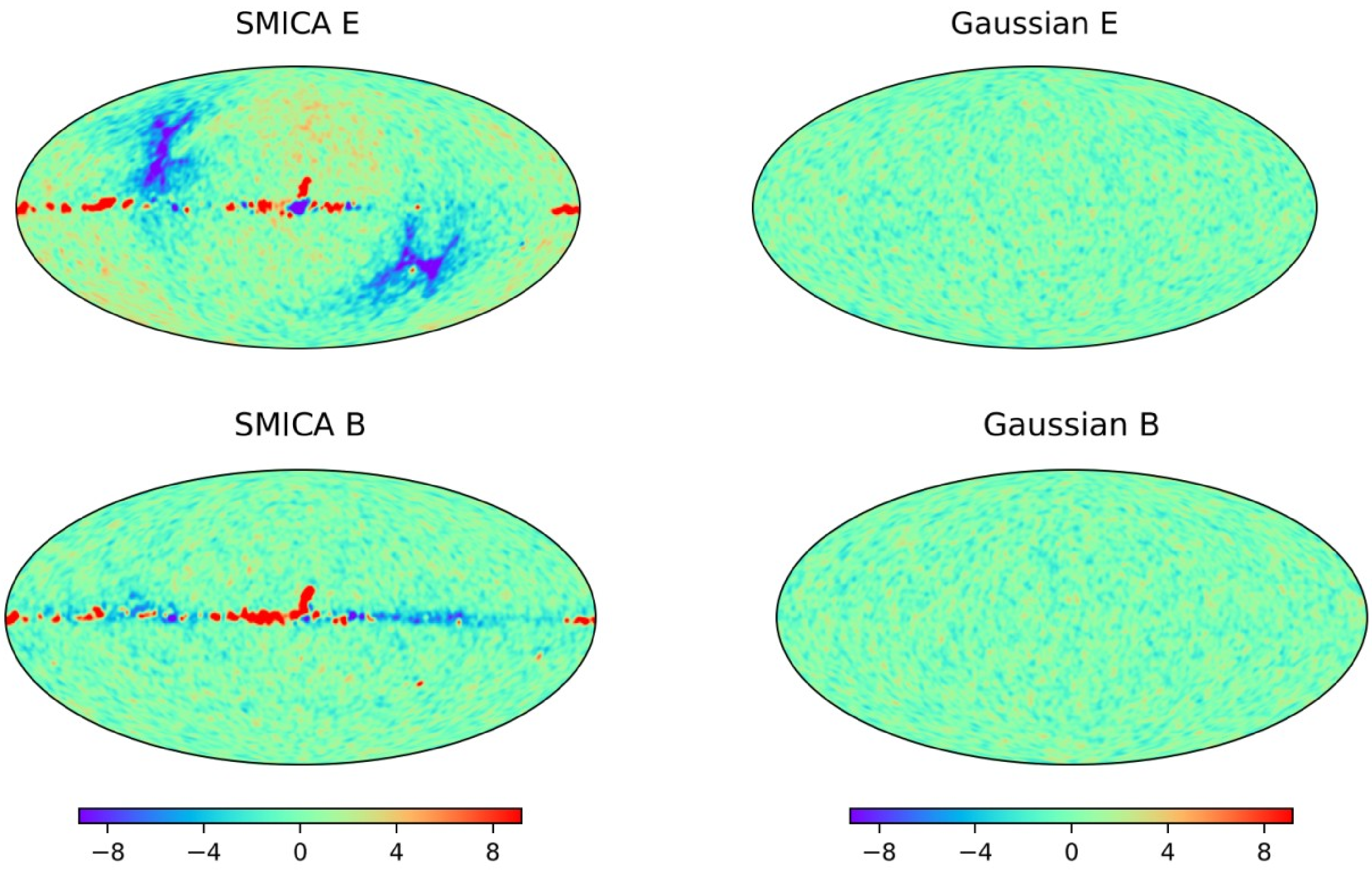
Fraction of points



Area fraction



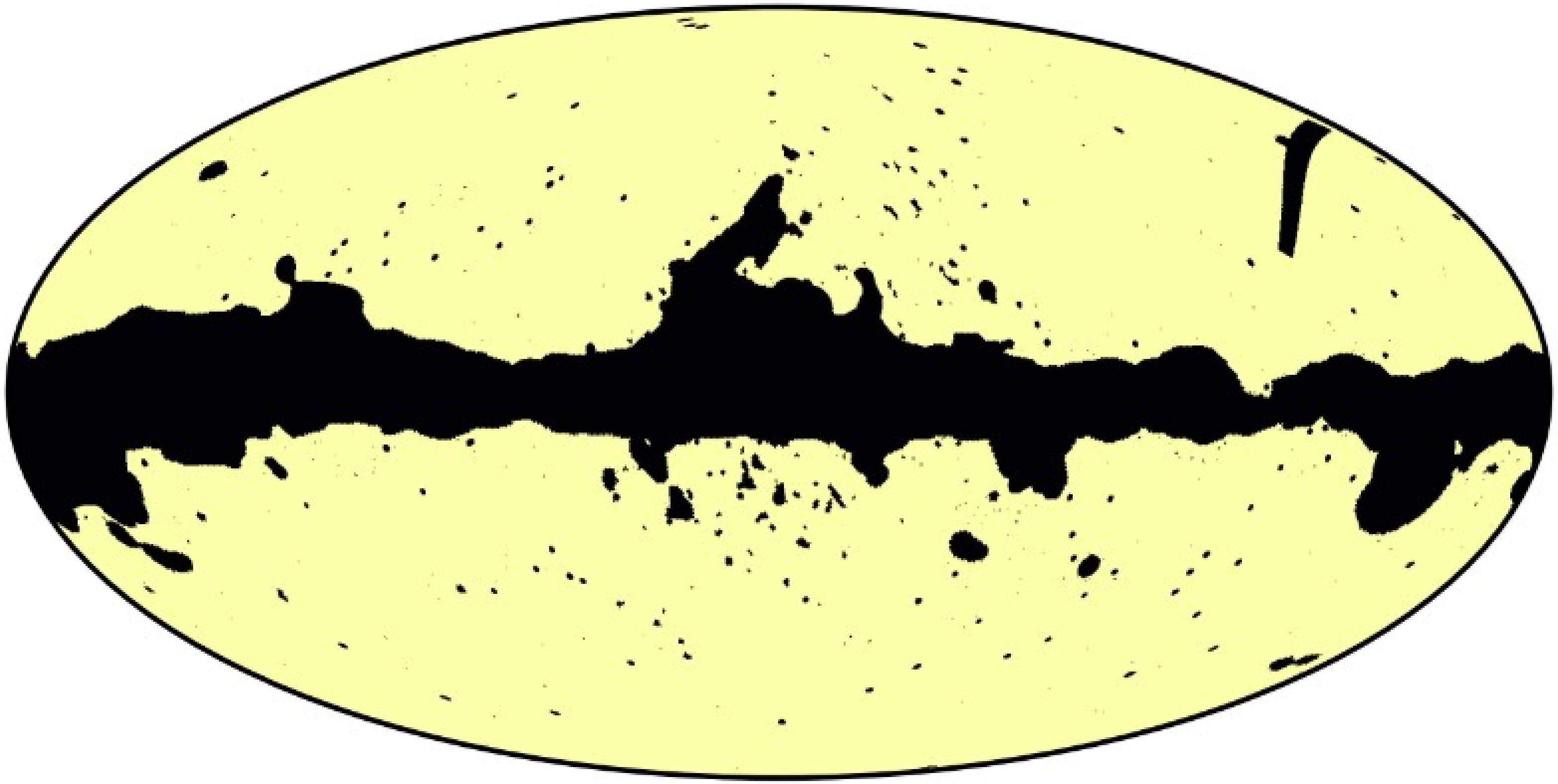
Planck data: density of unpolarized points



Novikov, Parfenov, Phys. Rev. D 111, 063536, arxiv.org/abs/2410.14351

Planck polarization data analysis:

Polarization confidence mask

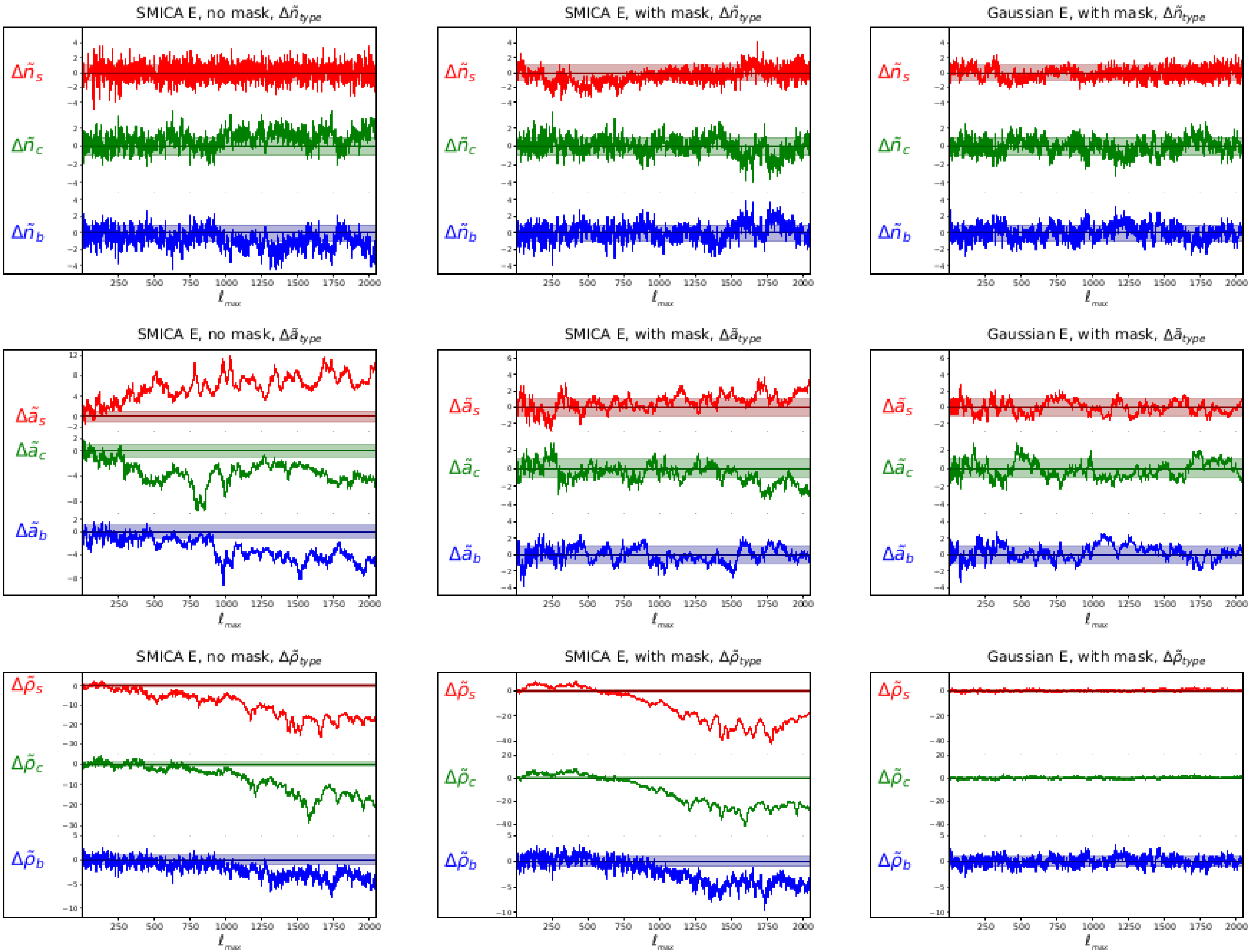


Planck E polarization data:

points

area

points / area



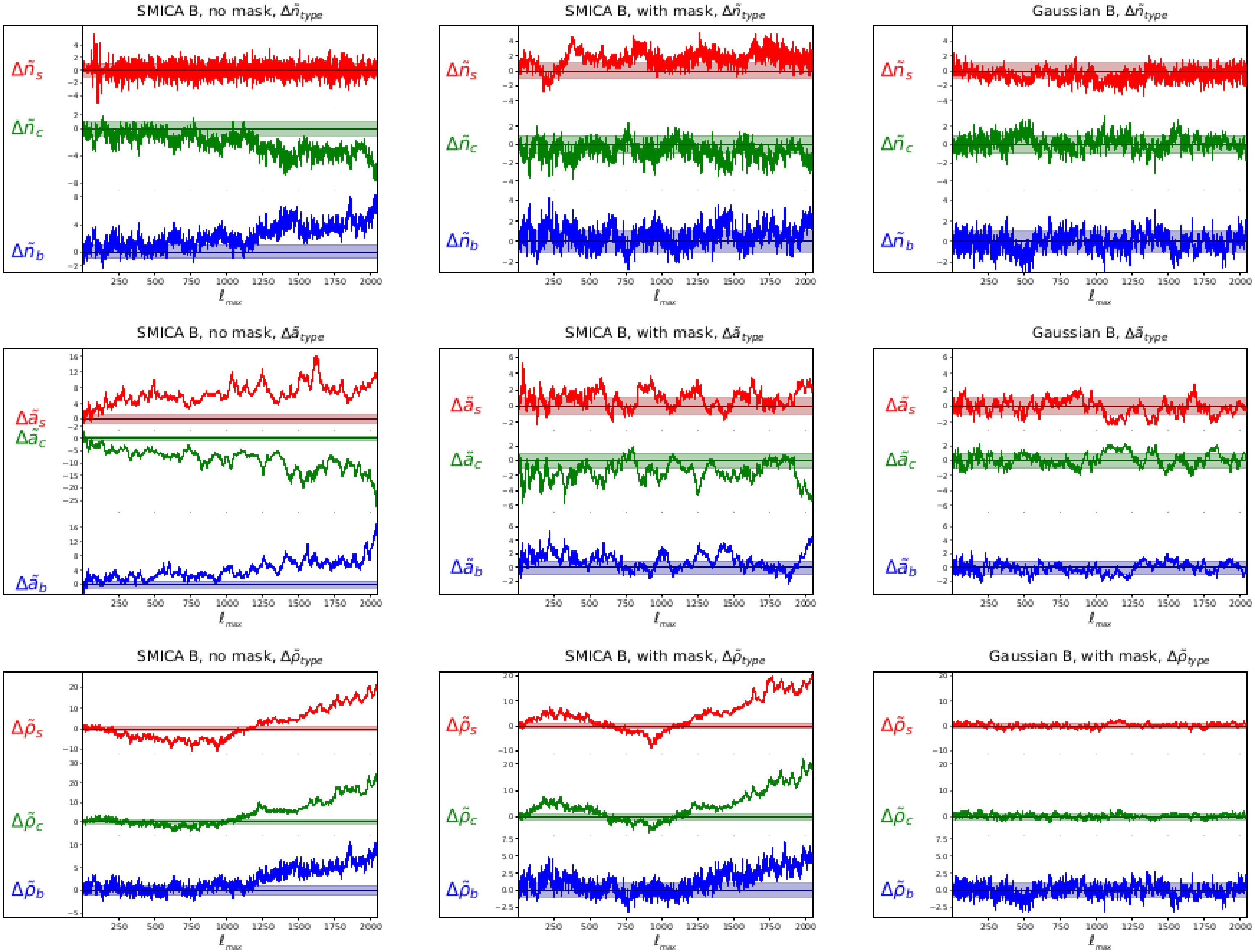
Parfenov, D.I. Novikov, A.O. Mihalchenko Phys. Rev. D 112, 103537, arxiv.2511.22220

Planck B polarization data:

points

area

points / area



Parfenov, D.I. Novikov, A.O. Mihalchenko Phys. Rev. D 112, 103537, arxiv.2511.22220

How to use this approach for SO and other CMB polarization data:

Available software:

https://github.com/koparfenov169-lab/CMB-polarization-Gaussianity-test

For any sky coverage
In HEALPix format

D. Novikov and K.Parfenov,, Phys. Rev. D 111, 063536, arxiv.org/abs/2410.14351

Problems with CMB data:

* Cosmic variance

* Quadrupole-octupole allignement

* Large scale anomalies

* Unremoved foregrounds

Prospects for independent measurement of

ℓ =1,2,3 CMB anisotropy multipoles using

the anisotropic Sunyaev-Zel'dovich effect

Anisotropic SZ effect (aSZ):

$$n_0(x) = B(x) = \frac{1}{e^x - 1}$$

$$\Theta_e = k_B T_e / m_e c^2$$

$$x = \frac{h\nu}{k_B T_r}$$

$$\Delta_n = n(x) - n_0(x)$$

Spectral distortion after scattering:

$$\begin{aligned}
\Delta_n/\tau &= \\
&= (\gamma_1 + \Theta_e \gamma_2) \left(-x \frac{dB}{dx}\right) + && \}1 \\
&+\Theta_e \gamma_3 \frac{1}{x^2} \frac{d}{dx} \left[x^4 \frac{d}{dx} \left(-x \frac{dB}{dx}\right)\right] + && \}2 \quad (aSZ) \\
&+\Theta_e \frac{1}{x^2} \frac{d}{dx} \left[x^4 \frac{dB}{dx}\right], && \}3 \quad (tSZ)
\end{aligned}$$

$$\gamma_3 = -\frac{3}{4\sqrt{\pi}} \left(\frac{4}{5\sqrt{3}} \tilde{a}_{10} - \frac{1}{3\sqrt{5}} \tilde{a}_{20} + \frac{1}{5\sqrt{7}} \tilde{a}_{30}\right)$$ → ℓ=1,2,3 anisotropy

$$\tau = \int N_e \sigma_T dr$$

I. Edigaryev, D. Novikov, and S. Pilipenko, Phys. Rev. D 98, 123513 (2018).

Chluba J., Nagai D., Sazonov S., Nelson K., 2012b, MNRAS, 426, 510, arXiv:1205.5778

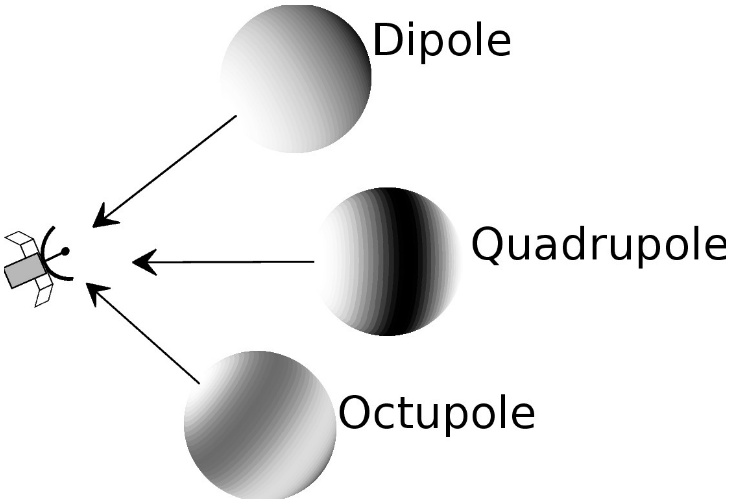
Anisotropic SZ effect (aSZ):

$$n_0(x) = B(x) = \frac{1}{e^x - 1} \qquad \Theta_e = k_B T_e / m_e c^2$$

$$x = \frac{h\nu}{k_B T_r} \qquad \Delta_n = n(x) - n_0(x)$$

Spectral distortion after scattering:

$$\begin{aligned}
&\Delta_n/\tau = \\
&= (\gamma_1 + \Theta_e \gamma_2)\left(-x\,\frac{dB}{dx}\right) + && \}1 \\
&+\Theta_e \gamma_3\, \frac{1}{x^2}\, \frac{d}{dx}\left[x^4 \frac{d}{dx}\left(-x \frac{dB}{dx}\right)\right] + && \}2 \quad (aSZ) \\
&+\Theta_e\, \frac{1}{x^2}\, \frac{d}{dx}\left[x^4 \frac{dB}{dx}\right], && \}3 \quad (tSZ)
\end{aligned}$$



$$\gamma_3 = -\frac{3}{4\sqrt{\pi}}\left(\frac{4}{5\sqrt{3}}\tilde{a}_{10} - \frac{1}{3\sqrt{5}}\tilde{a}_{20} + \frac{1}{5\sqrt{7}}\tilde{a}_{30}\right)$$ → ℓ=1,2,3 anisotropy

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tSZ:

$$I_{y_0}(\nu) = A_{y_0} x \frac{d}{dx}\left[x^4 \frac{dB}{dx}\right]$$

$$A_{y_0} = \frac{2(k_B T_r)^3}{(hc)^2} \tau \Theta_e$$

aSZ:

$$I_{asz}(\nu) = A_{y_0} x \frac{d}{dx}\left[x^4 \frac{d}{dx}\left(-x\frac{dB}{dx}\right)\right]\gamma_3$$

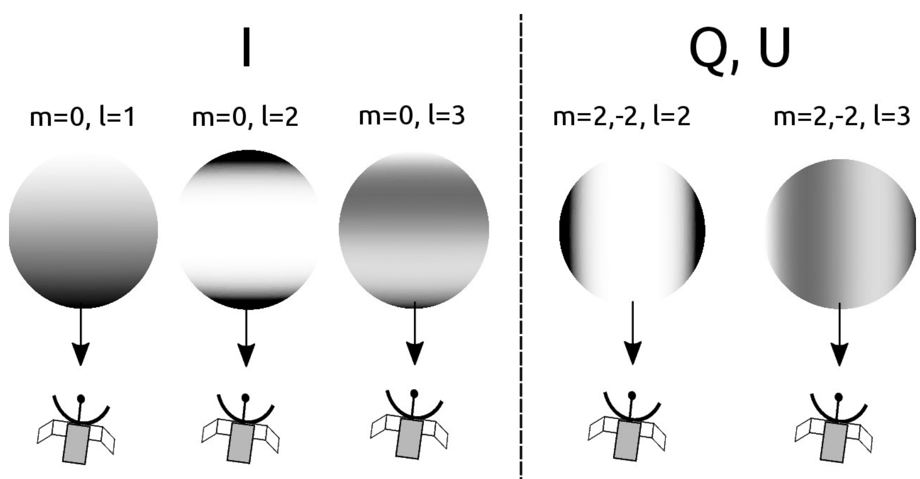
$$Q_{asz}(\nu) = A_{y_0} x \frac{d}{dx}\left[x^4 \frac{d}{dx}\left(-x\frac{dB}{dx}\right)\right]\gamma_3^q$$

$$U_{asz}(\nu) = A_{y_0} x \frac{d}{dx}\left[x^4 \frac{d}{dx}\left(-x\frac{dB}{dx}\right)\right]\gamma_3^u$$

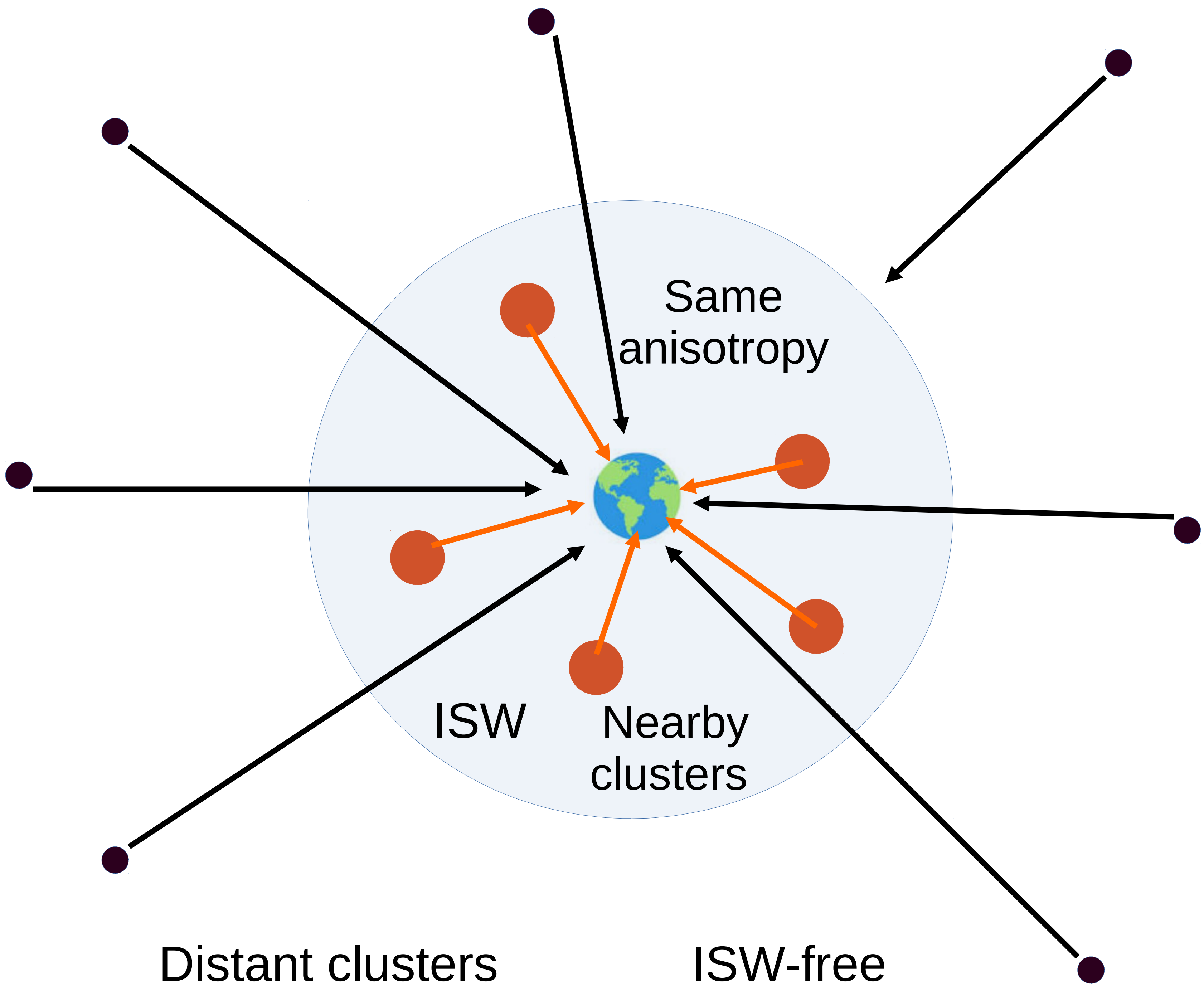
$$\gamma_3 = -\frac{3}{4\sqrt{\pi}}\left(\frac{4}{5\sqrt{3}}\tilde{a}_{10} - \frac{1}{3\sqrt{5}}\tilde{a}_{20} + \frac{1}{5\sqrt{7}}\tilde{a}_{30}\right)$$

$$\gamma_3^q = \frac{3}{4\sqrt{5\pi}}\left(\tilde{a}_{2,2} - \frac{1}{\sqrt{7}}\tilde{a}_{3,2}\right),$$

$$\gamma_3^u = \frac{3}{4\sqrt{5\pi}}\left(\tilde{a}_{2,-2} - \frac{1}{\sqrt{7}}\tilde{a}_{3,-2}\right)$$



Novikov, Pilipenko, De Petris, Luzzi & Mihalchenko, 2020, Physical Review D, 101(12), arXiv:2006.15571


Same
anisotropy
ISW
Nearby
clusters
Distant clusters
ISW-free

Requirements for the experiment:

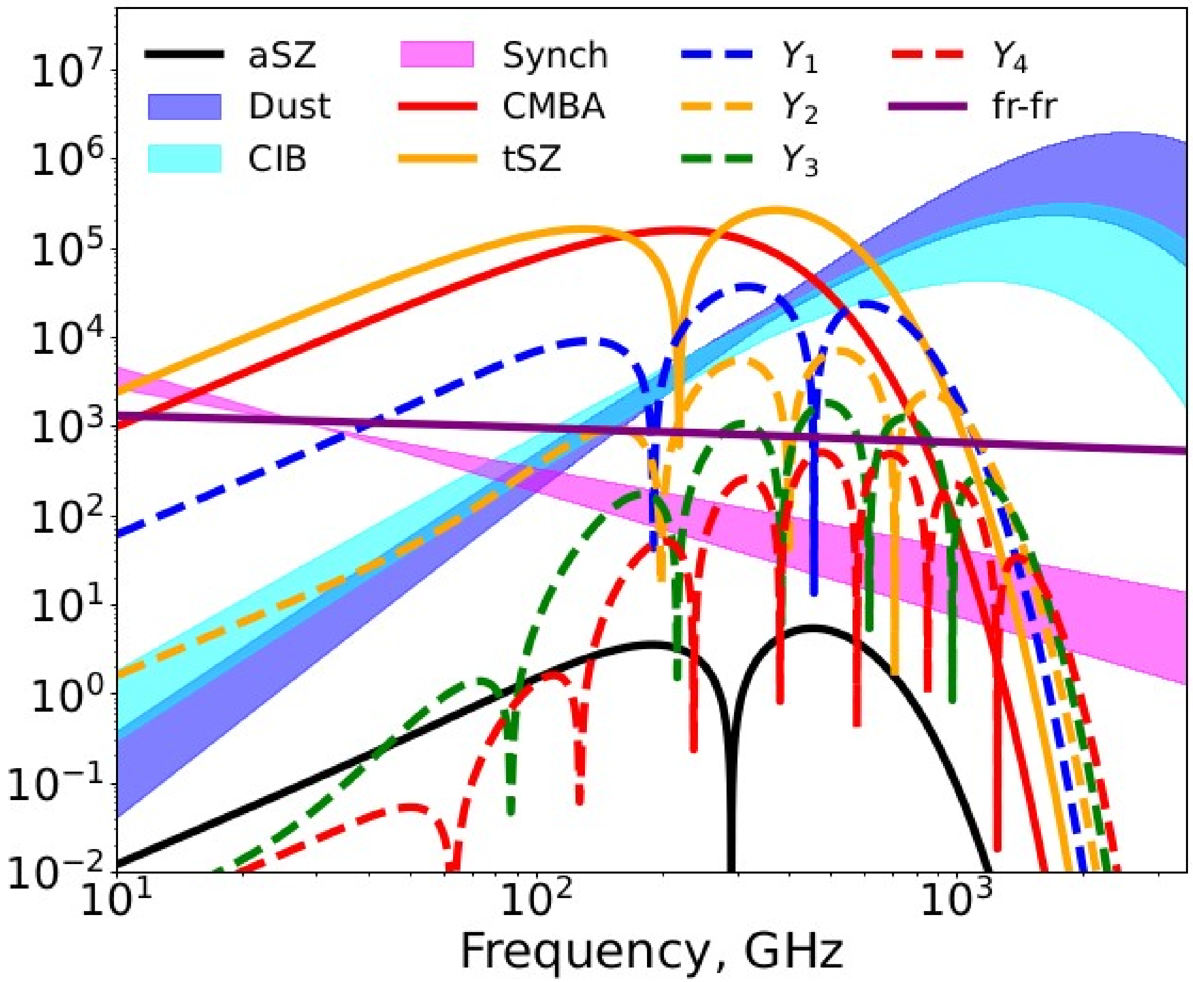
1. Space or Moon mission with cold (~10 K) large primary mirror;

2. Fourier Transform Spectrometer (FTS);

3. Frequency range 50-3000 GHz with a channel width ~10 GHz;

4. Special data cleaning.

Signal and foregrounds:



M. H. Abitbol, J. Chluba, J. C. Hill, and B. R. Johnson, MNRAS 471, 1126 (2017), 1705.01534.

D. Challinor, M. T. Ford, and A. N. Lasenby, MNRAS 312, 159 (2000)

D.I. Novikov et al, Phys. Rev. D, arxiv.org/abs/2512.07441

Data processing

Least Response Method (LRM):

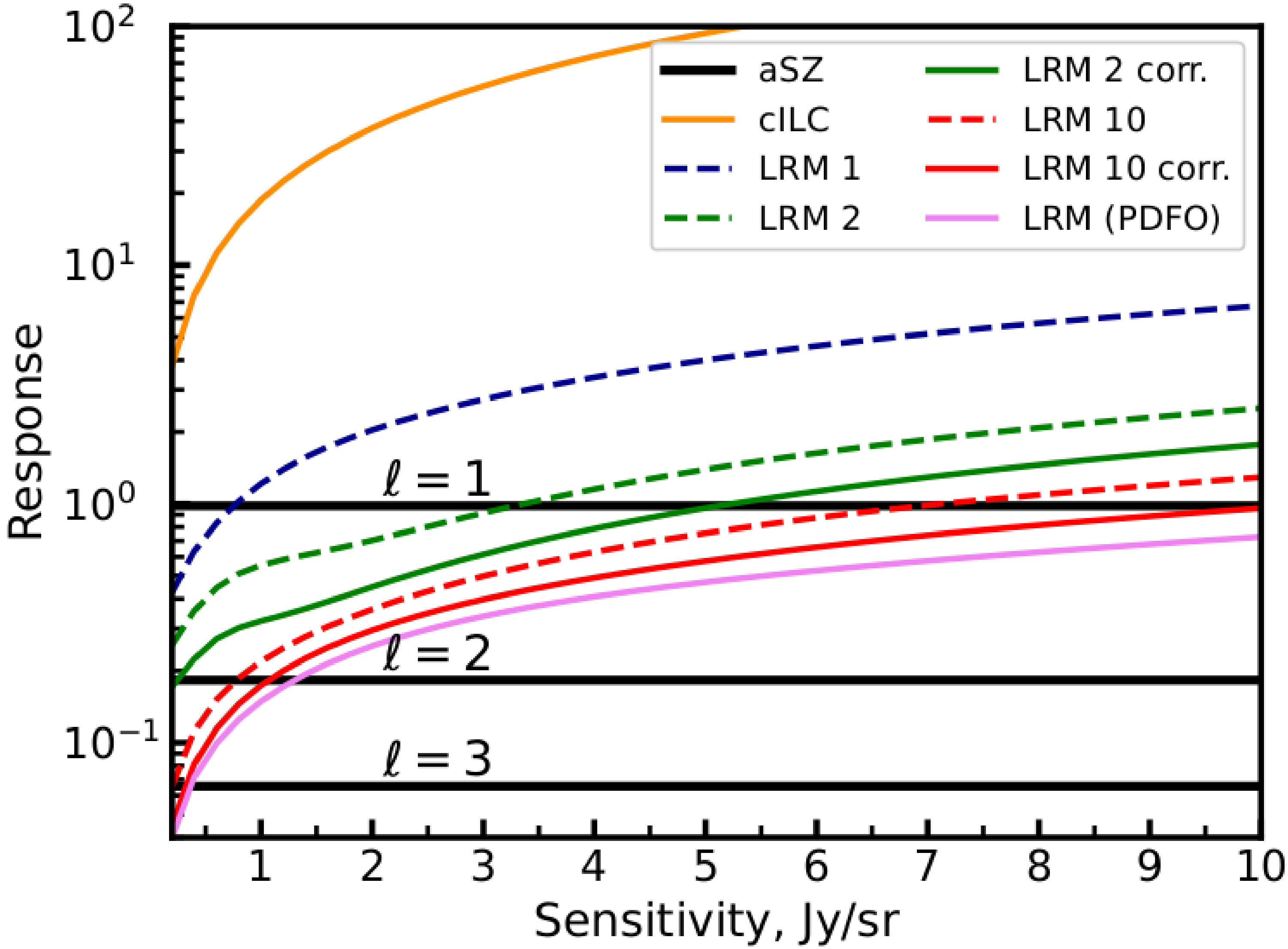
* It does not require complete orthogonalization of the various signal components.

* This approach significantly reduces the impact of photon noise on the recovered component amplitudes.

* It works much better than cILC or mILC.

Maillard, J.-P., Mihalchenko, A., Novikov, D., Osipova, A., Pilipenko, S., & Silk, J., 2024 Physical Review D, 109(2) , arXiv:2401.13415

Numerical results:



D.Novikov et al, Phys. Rev. D, arxiv.org/abs/2512.07441

Conclusions

1. The statistics of unpolarized points provides a highly sensitive Gaussianity test of CMB polarization across all angular scales:

1a. Planck polarization data are strongly non-Gaussian for both E and B modes.

1b. This approach can be applied to any data in HEALPix format using publicly available software: https://github.com/koparfenov169-lab/CMB-polarization-Gaussianity-test.

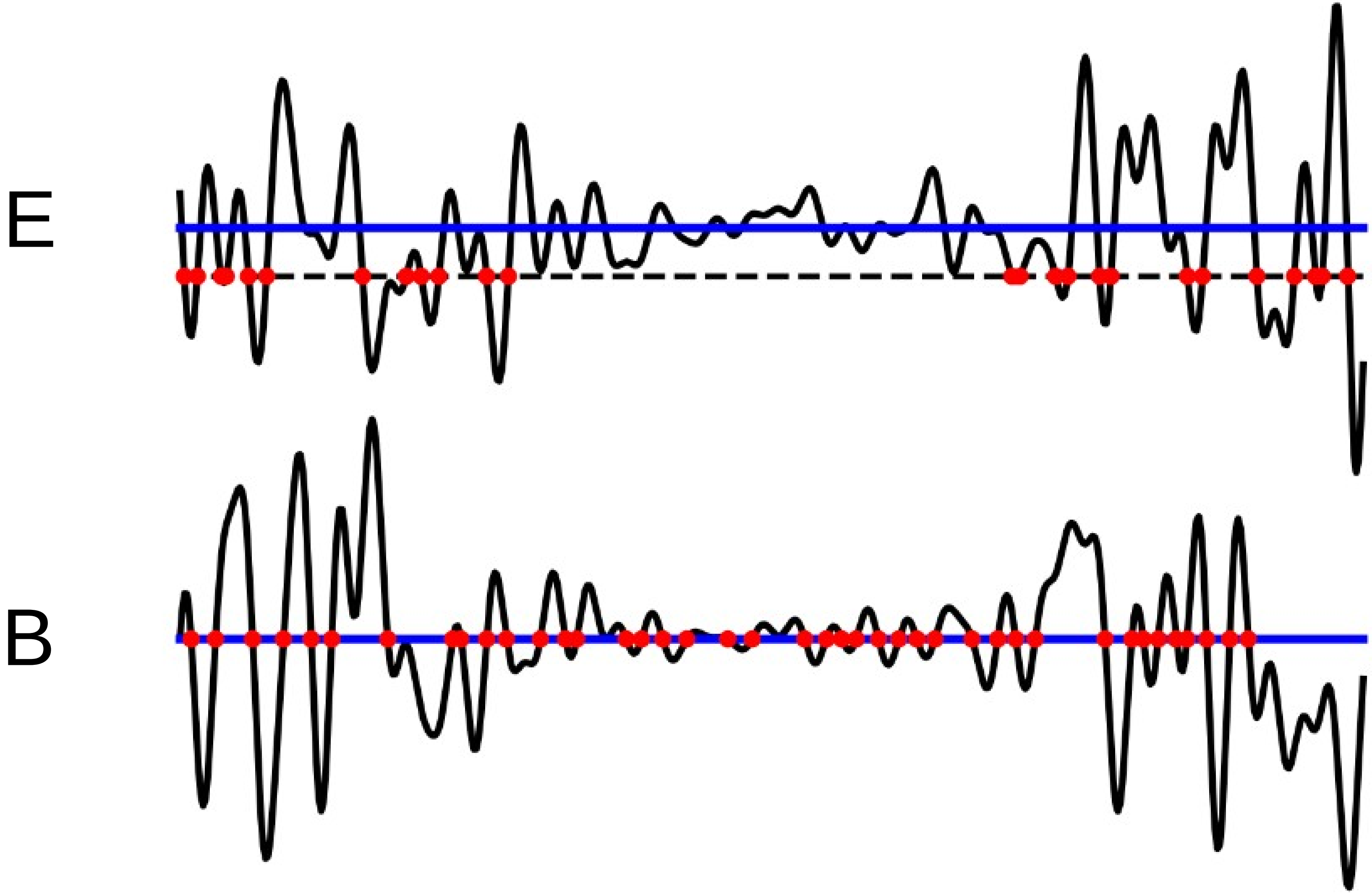
2. An alternative method for measuring CMB anisotropy using the aSZ effect makes it possible to:

2a. Measure the CMB dipole, quadrupole and octupole anisotropies at the locations of galaxy clusters;

2b. Separate the Sachs-Wolfe effect from the Integrated Sachs-Wolfe (ISW) effect and circumvent the cosmic variance problem;

2c. LRM approach reduces signal-to-noise ratio.

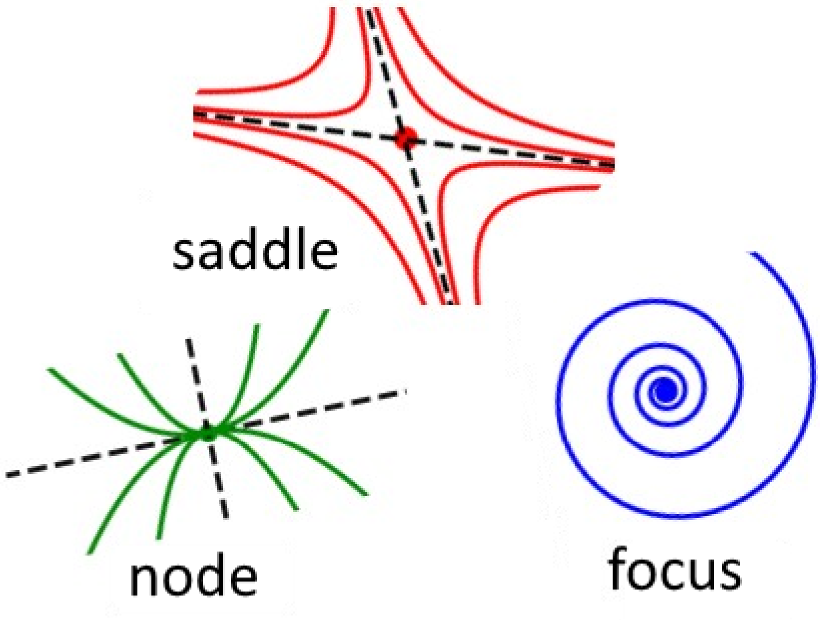
Thank you !!!



E
B

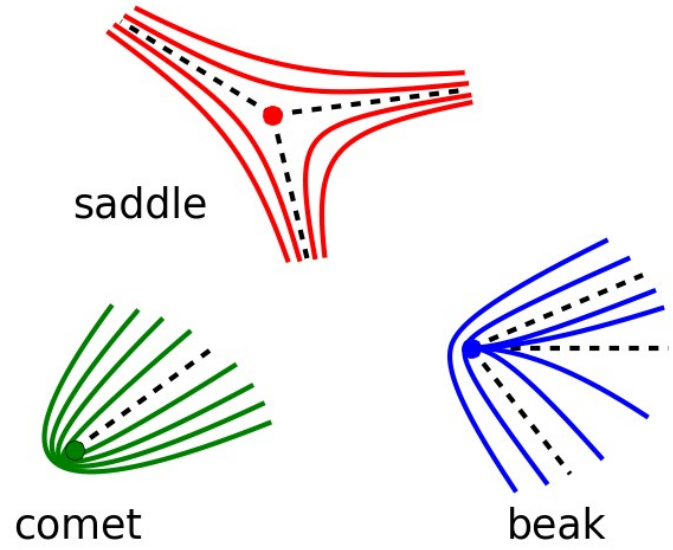
Zero points:

Vector field:



B≠0

Tensor field (polarization):



for any E or B